

MSTAR PRACTICE WORKSHEET

Lesson 3-10: Convert Measurement Units

Grade 6 Mathematics · Standard 6.AT.3c

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

How many inches are in 7 feet?

- ☐ A 70 inches
- ☐ B 72 inches
- ☒ C 84 inches
- ☐ D 96 inches

$7 \times 12 = 84$ inches.

Question 2 SELECTED RESPONSE — CORRECT: B

How many feet are in 3 yards?

- ☐ A 6 feet
- ☒ B 9 feet
- ☐ C 12 feet
- ☐ D 36 feet

1 yard = 3 feet. $3 \times 3 = 9$ feet.

Question 3 SELECTED RESPONSE — CORRECT: D

How many ounces are in 4 pounds?

- ☐ A 16 ounces
- ☐ B 32 ounces
- ☐ C 48 ounces
- ☒ D 64 ounces

1 pound = 16 ounces. $4 \times 16 = 64$ ounces.

Question 4 SELECTED RESPONSE — CORRECT: A

Convert 24 inches to feet.

- ☒ A 2 feet
- ☐ B 3 feet
- ☐ C 12 feet
- ☐ D 24 feet

$24 \div 12 = 2$ feet.

Question 5 SELECTED RESPONSE — CORRECT: B

A shelf is 30 inches long. How many feet and inches is that?

- ☐ A 1 foot 18 inches
- ☒ B 2 feet 6 inches
- ☐ C 2 feet 8 inches
- ☐ D 3 feet 0 inches

$30 \div 12 = 2$ remainder 6. So 30 inches = 2 feet 6 inches.

Question 6

SELECTED RESPONSE — CORRECT: D

Leg 1: The joystick rod is 3 feet long, but the parts catalog lists it in inches. How many inches is that? (1 ft = 12 in)

- ☐ A 15 inches
- ☐ B 30 inches
- ☐ C 33 inches
- ☒ D 36 inches

Inches are smaller than feet, so multiply: $3 \times 12 = 36$ inches.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: D

Convert 3 hours to seconds using 1 hour = 60 minutes and 1 minute = 60 seconds.

- ☐ A 120 seconds
- ☐ B 180 seconds
- ☐ C 3,600 seconds
- ☒ D 10,800 seconds

3 hours $\times$ 60 minutes per hour = 180 minutes. Then 180×60 seconds per minute = 10,800 seconds.

- **A:** That multiplies only two of the three numbers. Go hours to minutes to seconds: $3 \times 60 \times 60$.
- **B:** That is 3 hours converted to MINUTES. Convert once more: 180 minutes $\times$ 60 seconds per minute.
- **C:** That is the number of seconds in ONE hour. Multiply by 3 for three hours.

Part B — correct answer: D

In dimensional analysis, how are the conversion ratios arranged?

- ☐ A So all units are multiplied together and kept.
- ☐ B So the largest unit is always on top.
- ☐ C So the numbers are as small as possible.
- ☒ D So the unwanted units appear in both a numerator and a denominator and cancel.

Writing $3 \text{ hr} \times (60 \text{ min} / 1 \text{ hr}) \times (60 \text{ s} / 1 \text{ min})$ puts 'hr' and 'min' in both a numerator and a denominator, so they cancel and only seconds remain.

Question 8

SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about dimensional analysis:

- ☒ **A** The given measurement is written as a fraction over 1.
- ☒ **B** Conversion ratios are arranged so unwanted units cancel.
- ☒ **C** Only the unit that remains after cancelling appears in the answer.
- ☐ **D** Units may be ignored as long as the numbers are multiplied correctly.
- ☒ **E** More than one conversion ratio can be used in a single chain.

A, B, C, and E are true. D is false — tracking the units is exactly what makes the method work and reveals a ratio written upside down.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A recipe says you need 5 feet of ribbon, but the store only sells ribbon in inches. Another recipe says you need 2 yards of fabric, but your ruler measures in feet. Convert both measurements and explain the steps you used. Which conversion factor did you use each time?

Model answer: Both conversions go from a bigger unit to a smaller one, so both multiply. Using 12 inches per foot, $5 \times 12 = 60$ inches of ribbon; using 3 feet per yard, $2 \times 3 = 6$ feet of fabric. Naming the conversion factor each time is what explains why both numbers grew.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.